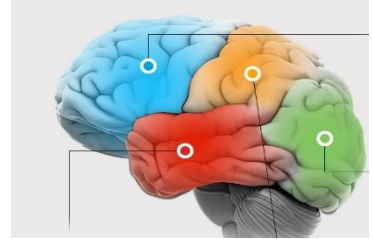


Pharmacotherapy of the Epilepsies

Seizure is a transient alteration of behavior due to the disordered, synchronous, and rhythmic firing of populations of brain neurons.

Epilepsy is a disorder of brain function characterized by the periodic and unpredictable occurrence of seizures.

seizures classified as:



partial seizures (focal seizures) originate from the temporal lobe

generalized seizures involve both hemispheres, tonic-clonic, absence, myoclonic.

Mechanism of action of antiepileptic drugs

- Prolong inactivation Na^+ channel that leads to block action potential propagation, stabilize neuronal membranes, and $\downarrow$ neurotransmitter release, focal firing, and seizure spread.

voltage or use dependent (greater effect if membrane is depolarized or used more)

- Ca^{2+} channel blockers
- $\uparrow$ inhibitory neurotransmitter (GABA) or GABA-mediated inhibition
- $\downarrow$ Fast excitatory neurotransmission (glutamate)

- To minimize toxicity, treatment with a single drug is preferred.
- If seizures are not controlled with the initial agent at adequate plasma concentrations, substitution of a second drug is preferred to the concurrent administration of another agent.
- However, multiple-drug therapy may be required, especially when two or more types of seizure occur in the same patient.

Antiepileptic drugs should be withdrawn slowly, not stopped abruptly.

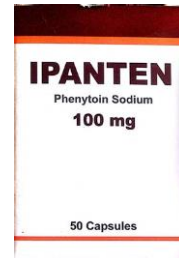
Phenytoin

Phenytoin is effective against focal and generalized tonic-clonic by slowing of the rate of recovery of voltage-activated Na⁺ channels from inactivation, voltage and use dependent.

Phenytoin is extensively bound (~90%) to serum proteins.

The rate of elimination of phenytoin varies. At low blood levels, metabolism follows first-order kinetics; as blood levels rise, the maximal limit of the liver to metabolize phenytoin is approached (zero order kinetics).

Phenytoin is used for trigeminal neuralgias



Drug Interactions

valproate co-administration resulting in marked increases in free phenytoin.

addition of phenytoin to a patient receiving ***warfarin*** can lead to bleeding disorders

phenytoin can ***induce the hepatic metabolizing enzymes.***

Fosphenytoin, a water-soluble prodrug for IV in the emergency treatment of status epilepticus. It is converted into phenytoin in liver and red blood cells.

Adverse Effects

- fosphenytoin can cause cardiac arrhythmias if administered at an excessive rate, the most notable toxic signs
- are with or without hypotension and CNS depression.
- chronic treatment with phenytoin causes gingival hyperplasia minimized by good oral hygiene.

- osteomalacia, and megaloblastic anemia. Hirsutism
- Hyperglycemia and glycosuria

Carbamazepine

It causes slowing of the rate of recovery of voltage-activated Na⁺ channels.

Carbamazepine is useful in patients with *generalized tonic-clonic and both focal, bipolar disorder, trigeminal and glossopharyngeal neuralgias*

carbamazepine has antidiuretic effects

Carbamazepine induces CYP



Adverse Effects

- Acute intoxication with carbamazepine can result in stupor or coma, hyperirritability, convulsions, and respiratory depression.
- During long term therapy, drowsiness, ataxia, diplopia, and blurred vision.
- serious hematological toxicity (aplastic anemia, agranulocytosis); and hypersensitivity reactions (skin reactions, eosinophilia, lymphadenopathy).
- A late complication of therapy with carbamazepine is retention of water, with decreased osmolality and concentration of Na⁺ in plasma, especially in elderly patients with cardiac disease.

Drug Interactions

It affects metabolism of other AEDs and affected by others

Oxcarbazepine

- Oxcarbazepine has a mechanism of action similar to that of carbamazepine

- but is a less-potent enzyme inducer than carbamazepine and does not induce the hepatic enzymes involved in its own degradation. Get attention of ?????
- Fewer hypersensitivity reactions have been associated with oxcarbazepine,
- hyponatremia may occur more commonly

Valproate

it is effective in inhibiting seizures in a variety of models (broad anticonvulsant).

effective in the treatment of absence, myoclonic, focal, and tonic-clonic seizures.

Mechanisms of Action

prolong recovery of voltage-activated Na⁺ channels from inactivation

Ca²⁺ currents effect on T-type currents

↑brain content of GABA,

(↑ synthesis inhibit & ↓degradation)



Adverse Effects and Drug Interactions

The most frequent side effects are transient **GI symptoms**, including anorexia, nausea, and vomiting (~16%).

CNS: sedation, ataxia, and tremor; decrease the dosage.

Rash, **hair fall and alopecia**

stimulation of appetite, **weight gain** has been seen with chronic treatment.

Elevation of hepatic transaminases in plasma is observed in up to 40% of patients and often occurs asymptotically during the first several months of therapy.

A rare but frequently fatal complication is **fulminant hepatitis**.

Children below 2 years of age with other medical conditions who were given multiple ASDs were especially likely to suffer fatal hepatic injury.

Acute pancreatitis and hyperammonemia frequently seen.

Valproate is a teratogenic effect, such as neural tube defects.

Benzodiazepines

enhance GABA-mediated synaptic inhibition. They act at GABA_A receptors and increase the frequency, but not duration, of openings at GABA-activated Cl⁻ channels

Clonazepam for absence and myoclonic seizures

Clorazepate is approved for focal seizures.

Midazolam for bouts of increased seizure

Diazepam and lorazepam for the status epilepticus.



Adverse Effects

The principal side effects are drowsiness and lethargy.

Barbiturates

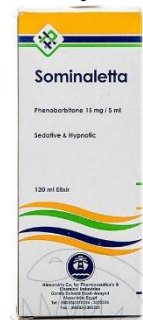
Phenobarbital was the first effective antiseizure agent. low toxicity, is inexpensive

phenobarbital increases the GABA_A receptor-mediated current by increasing the duration of opening. Phenobarbital induces CYP enzymes

Adverse

Sedation is the most frequent. irritability and hyperactivity in children and agitation and confusion in the elderly

Effects,



megaloblastic anemia that responds to folate and osteomalacia that responds to high doses of vitamin D occur during chronic phenobarbital therapy of epilepsy.

Drug Interactions

Concentrations of phenobarbital elevated by 40% during concurrent administration of valproate.

Ethosuximide

Ethosuximide is a primary agent for the treatment of generalized absence seizures.

Ethosuximide reduces low threshold T-type Ca²⁺ currents in thalamic neurons,

The most common dose-related side effects are GI complaints (nausea, vomiting, and anorexia) and CNS effects (drowsiness, lethargy, euphoria, and hiccough).

Levetiracetam

binding of levetiracetam to synaptic vesicle protein (SV2A) affect neuronal excitability by modifying the release of glutamate and GABA.

It is used for myoclonic, focal, and generalized tonic-clonic seizures in adults and children as young as 4 years old.

Levetiracetam is not bound to plasma proteins.



it is devoid of interactions with other ASDs, oral contraceptives, or anticoagulants.

Well tolerated. The most frequently reported adverse effects are somnolence, asthenia, ataxia, and dizziness. Behavioral and mood changes are serious,

Lamotrigine

It acts on Na⁺ channels and inhibits the release of glutamate



Lamotrigine is useful for monotherapy and add-on therapy of focal and secondarily generalized tonic-clonic seizures. *May used in pregnancy*

Concurrent use of lamotrigine and carbamazepine is associated with increases of the 10,11-epoxide of carbamazepine and clinical toxicity in some patients.

commonly dizziness, ataxia, blurred or double vision, nausea, vomiting, and rash

Stevens-Johnson syndrome and disseminated intravascular coagulation

Topiramate

Topiramate reduces Na⁺ currents, activates a hyperpolarizing K⁺ current, enhances postsynaptic GABA_A receptor currents, and ↓activation of the glutamate receptors.

The drug is a weak inhibitor of carbonic anhydrase.

The agent is effective as monotherapy for refractory focal epilepsy and refractory generalized tonic-clonic seizures.

Topiramate is well tolerated. The most common SE are somnolence, fatigue, weight loss, nervousness, cognitive impairment. It may precipitate renal calculi?

Vigabatrin

approved for refractory focal seizures and for treatment of infantile spasms

it increased concentrations of GABA in the brain, excreted unmetabolized renal

Due to progressive and permanent bilateral vision loss (FDA box warning), vigabatrin must be reserved for patients who have failed several alternative therapies. A patient's vision must be professionally monitored at the beginning of therapy and regularly throughout and after a therapeutic course.

Gabapentin and Pregabalin

they consist of a GABA molecule covalently bound to a lipophilic chain. after oral administration and are not metabolized and excreted unchanged, mainly in the urine (have no interactions with other ASDs).

They are effective for focal seizures, for the management of the neuropathic pain associated with postherpetic neuralgia in adults. pregnancy category C

Pregabalin is FDA-approved for fibromyalgia and the neuropathic pain associated diabetic peripheral neuropathy, or spinal cord injury.

Be a physician

Focal and Focal-to-Bilateral Tonic-Clonic Seizures

Carbamazepine and phenytoin. The choice depends on assessment of toxic effects.

Generalized Absence Seizures

Ethosuximide and valproate are equally effective. If tonic-clonic seizures are present or emerge during therapy, valproate is the agent of first choice.

lamotrigine is effective for newly diagnosed absence epilepsy

Myoclonic Seizures

Valproate is the drug of choice for myoclonic seizures. Levetiracetam for refractory generalized myoclonic seizures.

Febrile Convulsions

Between 2% and 4% of children experience a convulsion with a febrile illness; 25%–33% of these children will have another febrile convulsion. Only 2%–3% become epileptic in later years,

For children at high risk of developing recurrent febrile seizures and epilepsy, rectally administered diazepam at the time of fever may prevent recurrent seizures.

Seizures in Infants and Young Children

Infantile spasms are refractory to the usual ASD. Corticotropin or glucocorticoids are commonly used; Vigabatrin is efficacious

Status Epilepticus

It is a neurological emergency with mortality 20% . Parenteral Lorazepam, diazepam, phenytoin